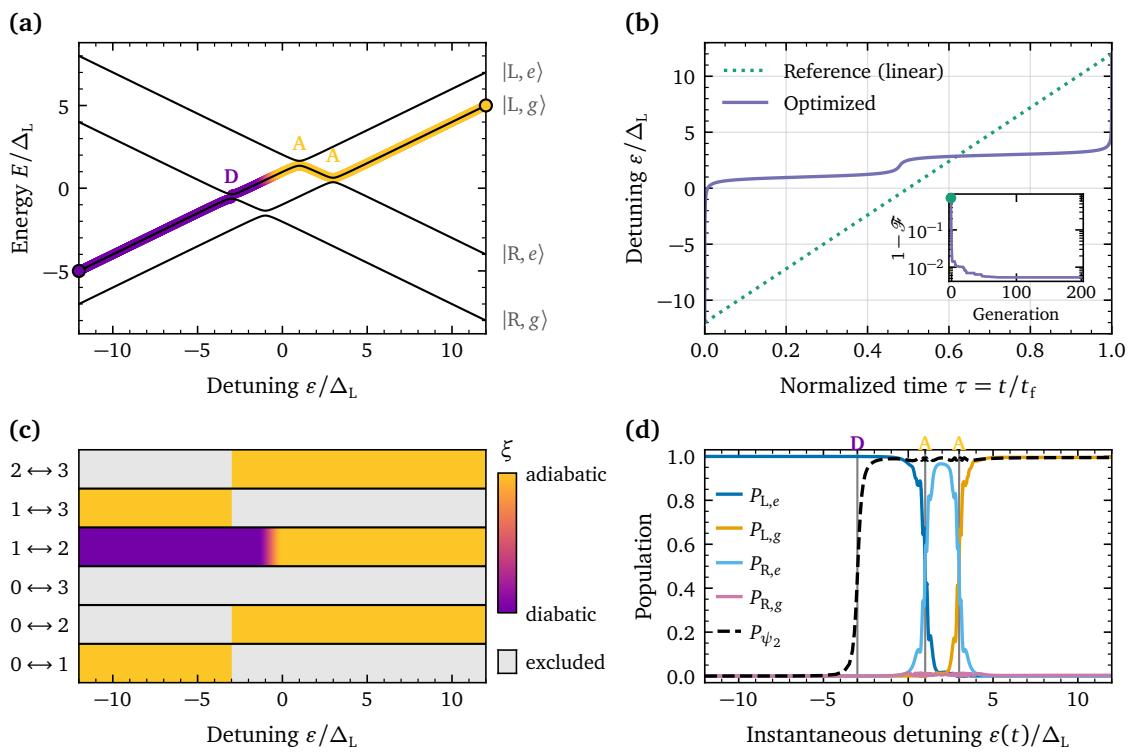

(a)
Energy E/Δ_L
Detuning ε/Δ_L
$|L,e\rangle$
$|L,g\rangle$
$|R,e\rangle$
$|R,g\rangle$
D
A
A
(b)
Detuning ε/Δ_L
Normalized time $\tau = t/t_f$
Reference (linear)
Optimized
$1-\mathcal{F}$
Generation
(c)
$2 \leftrightarrow 3$
$1 \leftrightarrow 3$
$1 \leftrightarrow 2$
$0 \leftrightarrow 3$
$0 \leftrightarrow 2$
$0 \leftrightarrow 1$
ξ
adiabatic
diabatic
excluded
Detuning ε/Δ_L
(d)
D
A
A
Population
$P_{L,e}$
$P_{L,g}$
$P_{R,e}$
$P_{R,g}$
P_{ψ_2}
Instantaneous detuning $\varepsilon(t)/\Delta_L$